

DSA-JC Computing / AI / Robotics / Leadership Portfolio

Leo Suite

AI for Smarter Schools and Student Support

Three deployed AI applications exploring student learning, campus robotics, and responsible wellbeing technology.

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Portfolio: leo-suite.vercel.app

GitHub: github.com/mentorkokkwa/leo-suite

Student Profile & Motivation

I am applying for DSA-JC with a focus on **Computing, AI, Robotics, and Leadership**. Rather than building isolated coding exercises, I designed **Leo Suite** — a coherent three-project portfolio that addresses real problems I observed in school: teachers overwhelmed by marking, growing interest in campus automation, and the risks of unstructured AI chat for student wellbeing.

My goal is to show not only technical ability, but also **product thinking, ethical AI design, and deployment discipline** — every project is live on Vercel with documented demo flows.

What I Built

Project	Domain	Core Skill Demonstrated
EduLens AI	AI for Smarter Schools	Structured LLM outputs, RAG, export pipelines
CampusBot AI	Robotics / Algorithms	A* pathfinding, simulation, metrics
YouthMentor AI	Leadership / Safety	Crisis guardrails, privacy-first UX

Technology Stack (Shared)

- Next.js 16
- React 19
- TypeScript
- Tailwind CSS 4
- Zod
- Vercel
- Multi-provider LLM

Why Three Apps?

A single bloated codebase would dilute the story. Three focused applications let reviewers evaluate depth in AI engineering (EduLens), algorithms (CampusBot), and safety ethics (YouthMentor) — while sharing a common TypeScript and deployment foundation.

Project Overview

Dimension	EduLens AI	CampusBot AI	YouthMentor AI
One-line value	Structured homework feedback & lesson generation for teachers and students	School robot simulator with A* navigation and task scenarios	Safety-first digital mentor for mood check-in and reflection
Primary users	Students, teachers, tutors	School IT, robotics club, DSA reviewers	Secondary students, counsellors (review)
Live demo	leo-suite-edutech-six.vercel.app/edulens	leo-suite-robot.vercel.app/campusbot	leo-suite-growth-swart.vercel.app/youthmentor
Public docs	leo-suite-edutech-showcase	leo-suite-robot (public)	leo-suite-growth-showcase
Data storage	Browser localStorage + in-memory API store	Client-side only (simulation)	Browser localStorage only — no server DB
Portfolio weight	Primary (4 pages)	Supporting (2 pages)	Supporting (2 pages)

EduLens AI — Problem & Users

Problem

Teachers spend 2–4 hours marking a single worksheet for 35 students. Students receive ticks and crosses but rarely get structured feedback on *why* they made mistakes or *what* to practise next. Generic AI chatbots give inconsistent free-form answers unsuitable for mistake books, dashboards, or printable reports.

Solution

EduLens returns **validated, structured JSON** — marking-style feedback, mistake type tags, similar practice questions, syllabus-aware lesson packs, and exportable PDF/DOCX/PPTX reports. Teachers remain in the loop via `teacherReviewRecommended` flags.

Target Users

- **Students (13–18):** Understand homework mistakes; build a mistake book
- **Teachers / tutors:** Draft lesson materials faster with syllabus RAG
- **DSA reviewers:** Evaluate AI EdTech safely via mock mode — no API keys exposed

EduLens AI — Core Features

1. Homework Analyzer

Text paste or image upload (vision OCR). Returns per-question scores, mistake types (*conceptual, calculation, careless, incomplete, method*), improvement tips, and similar questions.

Live: </edulens/homework-analyzer>

Upload a worksheet photo or paste text → structured feedback cards → mistake book entry

2. Lesson Generator

3-part generation pipeline with staged loading UX. Outputs lesson plan, worksheet, answer key, visual learning steps, and subject-aware diagrams. User selects deliverables (plan only, worksheet, etc.).

Live: </edulens/lesson-generator>

Enter topic + syllabus level → multi-stage progress → shareable report link

3. Mistake Book & Dashboard

Persistent local history groups errors by type and knowledge point. Dashboard aggregates usage stats for self-review.

4. Export Pipeline

Format	Use case
PDF	Printable homework report for parent-teacher conference
DOCX	Editable lesson plan for teacher refinement
PPTX	Classroom slides from visual lesson steps

EduLens AI — Architecture & Screenshots

High-level flow: React UI → Next.js API Routes → EduLens Service → Prompt Builder → Provider Chain → Zod Validator → UI + localStorage

API Routes (9 endpoints)

Route	Purpose
/api/edulens/homework/analyze	Run homework analysis
/api/edulens/lesson/generate	Generate lesson pack (multi-part)
/api/edulens/extract-content	Vision OCR
/api/edulens/mistake-book	Mistake book CRUD
/api/edulens/dashboard	Usage stats
/api/edulens/ai-status	Provider health (demo verification)

RAG & Syllabus

Indexed MOE-aligned topic catalog with PDF chunks. RAG context injected into prompts; post-generation sanitization removes citation leaks from lesson output.

Demo Screenshots (Visit Live URLs)

Homework Results Structured JSON feedback cards with mistake tags	Lesson Pack Plan + worksheet + visual steps with LaTeX math
Mistake Book Aggregated error patterns over time	AI Status /api/edulens/ai-status — mock mode indicator

EduLens AI — Technical Details & Learning Value

Multi-Provider LLM Chain

Unified `completeJSON()` interface with ordered fallback: Agnes → Gemini → Groq → OpenRouter → Mock. Vercel demo uses `EDULENS_AI_MODE=mock` for zero-cost, deterministic seeded responses.

Structured JSON Pipeline

- 1. Build prompt with RAG context
- 2. LLM completion → extract JSON
- 3. Zod validate → apply safety defaults
- 4. Enrich (visual lessons, media, assessment alignment)
- 5. On failure: compact prompt → next provider → mock

Quality Engineering

- ~90 library modules under `src/lib/edulens/`
- Vitest unit tests: prompt builder, JSON parser, RAG retrieval, visual type detection
- RAG citation sanitization; subject-aware diagram rendering
- 3-part lesson generation respects token limits

What I Learned (EduLens)

Lesson	Application
Structure beats chat	Zod schemas enable mistake books, exports, dashboards
Mock mode is engineering	Reliable DSA demos without API key exposure
Teachers in the loop	Disclaimers + review flags, not "AI replaces marking"
Token limits are real	Multi-part generation + compact/minimal prompt fallbacks

CampusBot AI — Scenario & Algorithm

CampusBot simulates how a service robot could navigate a school campus — delivering worksheets, guiding visitors, or returning library books. It demonstrates algorithmic thinking and robotics interest without requiring hardware in the DSA review.

Task Scenarios

- **Worksheet delivery:** Robot picks up materials from staff room, navigates to classroom
- **Visitor guide:** Escort from reception to destination with obstacle avoidance
- **Library return:** Collect items and navigate back to library

A* Pathfinding Engine

Grid-based school map with walls, doors, and dynamic obstacles. A* finds optimal path → step simulation → on collision, dynamic re-plan → metrics report.

Algorithm Flow

1. Parse task goal cell from scenario definition
2. Build cost grid from map (walls = infinite cost)
3. Run A* with Manhattan heuristic
4. Simulate step-by-step movement
5. On blocked cell: invalidate path segment, re-plan from current position
6. Output metrics: total steps, collisions, re-plans, success/fail

Live: /campusbot/simulator

Select scenario → watch robot navigate → view metrics report

CampusBot AI — Metrics & Technical Evidence

Simulation Metrics

Metric	What it shows
Steps taken	Path efficiency vs optimal A* distance
Collisions	Dynamic obstacle handling effectiveness
Re-plans	Robustness when environment changes mid-task
Task completion	End-to-end scenario success rate

Tech Stack

Next.js

TypeScript

Client-side simulation

Canvas / SVG map

No API keys required

What I Learned (CampusBot)

- Pathfinding algorithms have direct real-world application in robotics and logistics
- Simulation before hardware reduces cost and accelerates iteration
- Clear metrics make algorithm behaviour reviewable by non-coders (teachers, reviewers)

Future: Hardware Bridge

The simulator is designed as a software prototype for a future Arduino / ROS2 hardware integration — same task definitions, real motor control. Planned as JC project extension.

School Map View

Grid layout with rooms, corridors, obstacles — robot path highlighted in real time

YouthMentor AI — Safety-First Wellbeing Design

YouthMentor addresses student exam stress and emotional pressure — but unlike engagement-optimised AI companions, it is designed with **hard safety boundaries** and **structured reflection** rather than open-ended chat.

Core Flow

- 1. Mood check-in + stress source selection (+ voice input option)
- 2. Choose mentor persona
- 3. Six guided reflection prompts (CBT-inspired framework)
- 4. **Safety pipeline** runs before any coaching response
- 5. Normal risk → mentor JSON response + action plan
- 6. High risk → crisis panel only (helplines, no coaching LLM)

Safety design principle: Never rely on the LLM alone for crisis detection. Keyword scan runs first; high-risk input blocks before any coaching prompt is sent.

Safety Pipeline

Layer	Action
1. Keyword scan	Detect self-harm / crisis language
2. Risk classifier	Score remaining input
3. Crisis-only UI	Helplines and resources — zero coaching LLM

Safety walkthrough: leo-suite-growth-swart.vercel.app/youthmentor/safety-demo

YouthMentor AI — Features & Responsible Design

Additional Features

- **Mood trend chart:** Visualise intensity from saved reflection history
- **Voice input:** Web Speech API on check-in, reflection, character chat
- **Trust metrics:** Home page compares responsible design vs engagement-first AI companions
- **Character chat:** Empathy-first protocol; labeled AI simulation (not real persons)
- **Privacy:** All data in browser localStorage — no account, no cloud storage

What YouthMentor is NOT

- Not therapy, diagnosis, or emergency care
- Not official communication from any real person or faith leader
- Not an engagement-optimised companion designed to maximise chat volume

What I Learned (YouthMentor)

Lesson	Evidence in product
Safety cannot be an afterthought	3-layer pipeline before coaching LLM
Structure reduces harm	6-prompt framework vs infinite open chat
Privacy builds school trust	localStorage only; no server database
Empathy ≠ dependency	Empathy-first replies + real-world help-seeking prompts

Crisis Panel Demo

Pre-loaded high-risk sample → guardrail blocks coaching → crisis resources only

System Architecture & Deployment

Deployment Topology

Vercel Project	App	Production URL
leo-suite	Portfolio home (this site)	leo-suite.vercel.app
leo-suite-edutech	EduLens AI	leo-suite-edutech-six.vercel.app
leo-suite-growth	YouthMentor AI	leo-suite-growth-swart.vercel.app
leo-suite-robot	CampusBot AI	leo-suite-robot.vercel.app

Repository Strategy

Full source for EduLens and YouthMentor lives in **private** deployment repos (API routes, prompts). **Public showcase** repos contain documentation, demo links, and non-sensitive type samples — protecting IP while enabling DSA portfolio review.

Shared AI Layer

EduLens and YouthMentor share LLM abstraction via EDULENS_* environment variables. Provider chain with mock fallback ensures reliable demos on Vercel without API keys.

Demo Links & GitHub

Resource	URL
Portfolio home	https://leo-suite.vercel.app
Meta repo	https://github.com/mentorkokkwa/leo-suite
EduLens showcase	https://github.com/mentorkokkwa/leo-suite-edutech-showcase
YouthMentor showcase	https://github.com/mentorkokkwa/leo-suite-growth-showcase
CampusBot source	https://github.com/mentorkokkwa/leo-suite-robot

Future Roadmap & School Contribution

Planned Extensions

Project	Next step	School value
EduLens	Teacher pilot for lesson prep; class dashboard	Reduce marking prep time
CampusBot	Arduino / ROS2 hardware prototype	Robotics club, STEM showcase
YouthMentor	Counsellor review of safety demo; anonymous insights	Wellbeing support tool evaluation

How I Can Contribute to School

- Lead AI literacy workshops — responsible use, not just prompt engineering
- Support teachers exploring AI-assisted lesson preparation (EduLens demo)
- Contribute to robotics / computing club with CampusBot simulator
- Advocate for safety-first youth AI design based on YouthMentor research

Commercial & Social Thinking

Freemium model (20 msgs/day) for YouthMentor; school B2B anonymous insights (no chat transcripts, PDPA-minded). EduLens positioned as teacher-augmenting, not replacing professional judgment.

Thank you for reviewing Leo Suite. All projects are live — please click the demo links and explore.
For questions: github.com/mentorkokkwa